

A new hyperchaotic system and its circuit implementation

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ABSTRACT

In this paper, a new hyperchaotic system is presented by adding a nonlinear controller to the three-dimensional autonomous chaotic system. The generated hyperchaotic system undergoes hyperchaos, chaos, and some different periodic orbits with control parameters changed. The complex dynamic behaviors are verified by means of Lyapunov exponent spectrum, bifurcation analysis, phase portraits and circuit realization. The Multisim results of the hyperchaotic circuit were well agreed with the simulation results.

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1. Introduction

Hyperchaos was firstly reported by Rössler in 1979 [1], and the first circuit implementation of hyperchaos was realized by Matsumoto et al. [2]. Hyperchaotic system is usually defined as a chaotic system with more than one positive Lyapunov exponent, indicating that the chaotic dynamics of the system are expanded in more than one direction giving rise to a more complex attractor. Hyperchaos has been studied with increasing interest in recent years, in the fields of nonlinear circuits [3], secure communications [4,5], lasers [6], Colpitts oscillators [7], control [8–11], and synchronization [12–15]. Due to its great potential in technological applications, the generation of hyperchaos has become a focal topic for research recently [16–20], in particular purposefully designing a hyperchaotic system from an originally chaotic but non-hyperchaotic system with some simple feedback control techniques, is a theoretically very attractive task. For example, several four-dimensional hyperchaotic systems [16–20] have been found in this way, based on some well-known three-dimensional chaotic system such as on the Chen system [21], the Lü system [22], the generalized Lorenz system [23], a unified chaotic system [24], and so on. This, however, is technically challenging, due to its very complicated hyperchaotic behavior and the lack of a general fundamental theory.

This Letter presents a new hyperchaotic system, which is generated by driving the three-dimensional autonomous chaotic system [25] with a nonlinear controller. The generated hyperchaotic system is not only demonstrated by Lyapunov exponent spectrum, bifurcation analysis, phase portraits but also verified with circuit realization. The Multisim results of the hyperchaotic circuit show very good agreement with the simulation results. For this three-dimensional autonomous chaotic system, we give some discussions on how to build the control law so that hyperchaoticity be observed. It may be the inspiration to the formation and perfection of the general control law.

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2. Design of new hyperchaotic system

Qi et al. introduced a new three-dimensional autonomous chaotic system [25], which is described as

$$\begin{cases} \dot{x} = a(y - x) + yz \\ \dot{y} = cx - y - xz \\ \dot{z} = xy - bz \end{cases}, \quad (1)$$

where x, y, z are state variables and a, b, c are system parameters. When $a = 35, b = 8/3, c = 55$, system (1) exhibits a chaotic behavior with a positive Lyapunov exponent. The corresponding chaotic attractor is depicted in Fig. 1. More detailed complex dynamics of system (1) can be seen in Ref. [25].

We know that, in order to obtain hyperchaos, two important requisites are as follows:

- (1) The minimal dimension of the phase space that embeds a hyperchaotic attractor should be at least four, which requires the minimum number of coupled first-order autonomous ordinary differential equations to be four.
- (2) The number of terms in the coupled equations giving rise to instability should be at least two, of which at least one should have a nonlinear function [1].

Now, by introducing a simple dynamic feedback control term w to the second equation of system (1), the new dynamic system is obtained:

$$\begin{cases} \dot{x} = a(y - x) + yz \\ \dot{y} = cx - y - xz + w \\ \dot{z} = xy - bz \\ \dot{w} = -xz + rw \end{cases}, \quad (2)$$

where r is a control parameter, determining the chaotic attractor and bifurcations of system (2). Obviously, the chaotic system (2) is a four-dimensional dynamical system, which has four Lyapunov exponents. It will be shown that this new system can be hyperchaotic with some suitably chosen control parameter r .

To ensure system (2) be dissipative, it is required that

$$\nabla V = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z} + \frac{\partial \dot{w}}{\partial w} = -a - 1 - b + r = r - 38.667 < 0. \quad (3)$$

Therefore, theoretically the largest value of r is 38.667. Choose r near zero so that the previous dissipative structure can be maintained most, according to the method presented by Ramasubramanian and Sriram [26], we obtain when $r = 1.3$, the Lyapunov exponents: $\lambda_1 = 1.4164, \lambda_2 = 0.5318, \lambda_3 = 0, \lambda_4 = -39.1015$. It is obvious that system (2) exhibits a hyperchaotic behavior. The projections of the hyperchaotic attractor are shown in Fig. 2.

3. Bifurcation analysis

Due to the lack of systematic methodology for purposefully designing a hyperchaotic system to date, the following investigation relies on a combination of mathematical analysis and numerical simulations. In system (2), when parameter r varies, several simulations are carried out, and the outcome of chaotic attractors and careful bifurcation analysis is summarized. Assume the Lyapunov exponents of system (2) are $\lambda_1, \lambda_2, \lambda_3$ and λ_4 . We found that:

- (1) For periodic orbits, $\lambda_1, \lambda_2, \lambda_3 < 0, \lambda_4 = 0$.

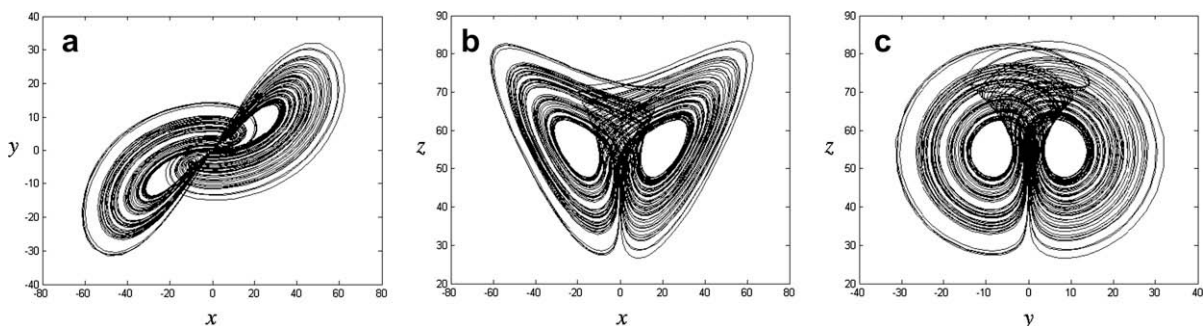


Fig. 1. The projections of attractor system (1).

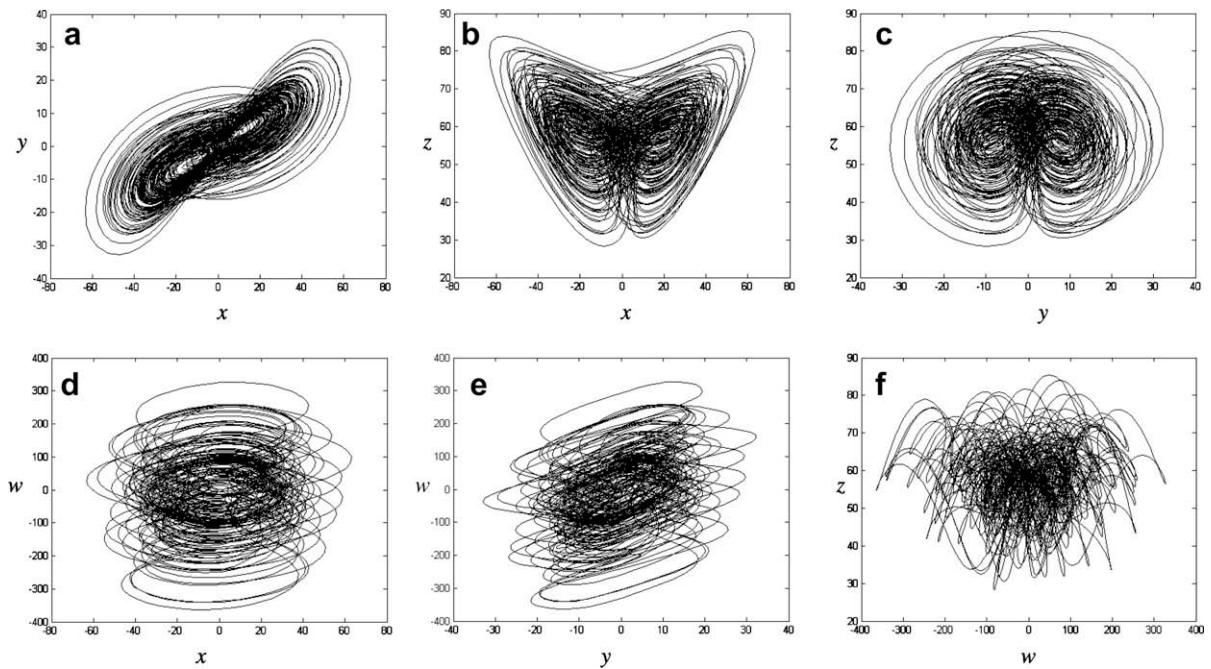


Fig. 2. When $r = 1.3$, the projections of attractor of system (2).

- (2) For chaotic attractor, $\lambda_1 > 0$, $\lambda_2, \lambda_3 < 0$, $\lambda_4 = 0$.
- (3) For hyperchaotic attractor, $\lambda_1, \lambda_2 > 0$, $\lambda_3 < 0$, $\lambda_4 = 0$.

According to the method presented by Ramasubramanian and Sriram [26], we obtain the Lyapunov exponents spectrum of system (2) in Fig. 3. As r varies, the stability of the system (2) is summarized as follows:

- (1) When $-11.19 < r \leq -10.64$, $-7.35 \leq r \leq -7.21$ or $-7.19 \leq r \leq -4.05$ is satisfied, $\lambda_1, \lambda_2, \lambda_3 < 0$, $\lambda_4 = 0$, the new system (2) undergoes periodic orbits.
- (2) When $-12 \leq r \leq -11.19$, $-10.64 < r < -7.35$, $-7.21 < r < -7.19$ or $-4.05 < r \leq 0.41$ is satisfied, $\lambda_1 > 0$, $\lambda_2, \lambda_3 < 0$, $\lambda_4 = 0$, the new system (2) is chaotic.
- (3) When $0.41 < r \leq 3$ is satisfied, $\lambda_1, \lambda_2 > 0$, $\lambda_3 < 0$, $\lambda_4 = 0$, the new system (2) is hyperchaotic.

The phase portraits of the system (2) with different r are also shown in Fig. 4.

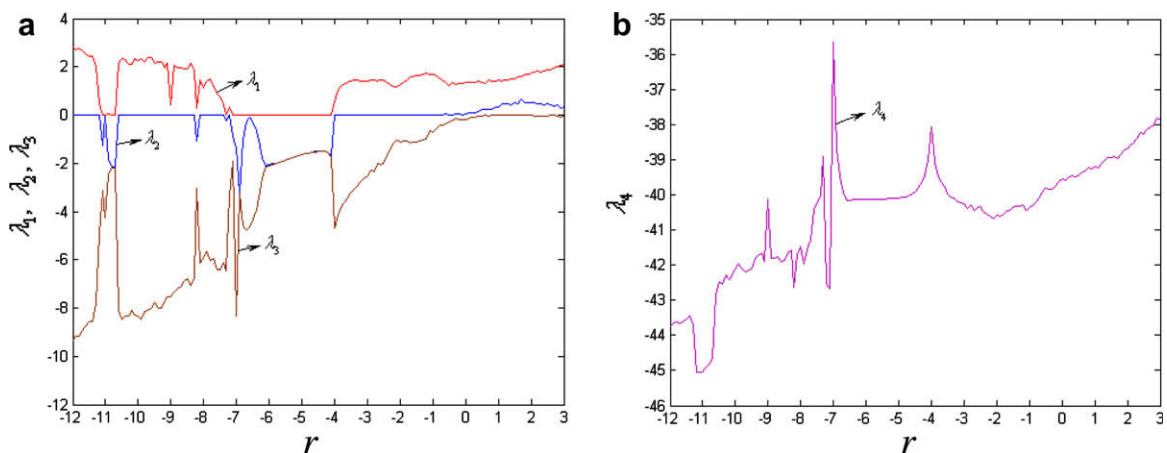


Fig. 3. Lyapunov exponents spectrum for $-12 \leq r \leq 3$.

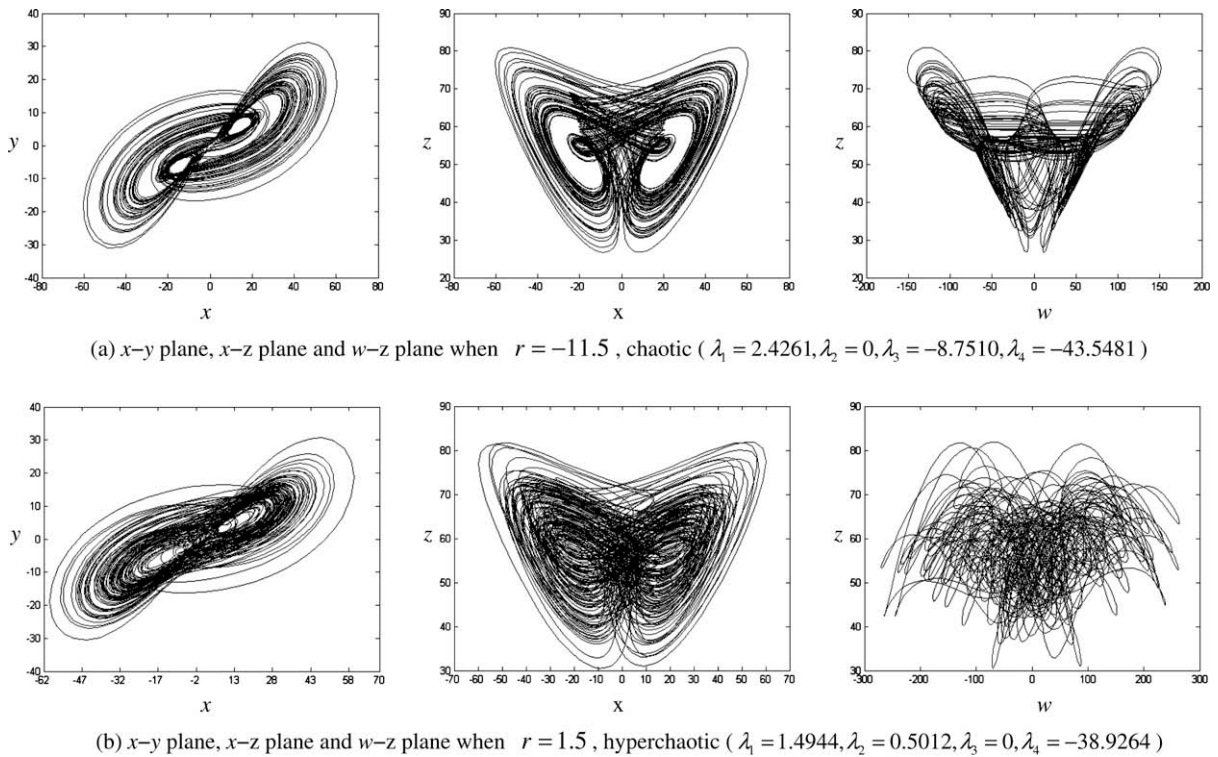


Fig. 4. Phase portraits of the system (2) for different r .

4. Circuit design of the hyperchaotic system

Considering the voltage restraint of practical electronic components, let

$$x = 10x', \quad y = 10y', \quad z = 10z', \quad w = 10w'. \quad (4)$$

Because the replacement of system variables does not influence system states and properties, we let

$$x = x', \quad y = y', \quad z = z', \quad w = w'. \quad (5)$$

then in the new coordinate system, the reduced hyperchaotic system (2) will be described as

$$\begin{cases} \dot{x} = a(y - x) + 10yz \\ \dot{y} = cx - y - 10xz + w \\ \dot{z} = 10xy - bz \\ \dot{w} = -10xz + rw \end{cases} \quad (6)$$

The designed circuit diagram of system (6) is shown in Fig. 5.

In Fig. 5, the voltages of C_1, C_2, C_3, C_4 are used as x, y, z and w , respectively. The operational amplifiers and associated circuitry perform the basic operations of addition, subtraction, and integration. The nonlinear terms of system (6) are implemented with the analog multipliers. The corresponding circuit equation can be described as

$$\begin{cases} \dot{x} = \frac{R_4}{R_1 R_5 C_1} y - \frac{R_4 R_7}{R_3 R_6 R_5 C_1} x + \frac{R_4}{R_2 R_5 C_1} yz \\ \dot{y} = \frac{R_{11}}{R_8 R_{12} C_2} x - \frac{R_{11} R_{14}}{R_{15} R_{13} R_{12} C_2} y - \frac{R_{11} R_7}{R_9 R_6 R_{12} C_2} xz + \frac{R_{11}}{R_{10} R_{12} C_2} w \\ \dot{z} = \frac{R_{18}}{R_{16} R_{19} C_3} xy - \frac{R_{18} R_{21}}{R_{17} R_{20} R_{19} C_3} z \\ \dot{w} = -\frac{R_{23} R_7}{R_{22} R_6 R_{24} C_4} xz + \left(\frac{R_{23}}{R_{27} R_{24} C_4} - \frac{R_{23} R_{26}}{R_{28} R_{25} R_{24} C_4} \right) w \end{cases} \quad (7)$$

We choose $R_1 = R_3 = 4k\Omega$, $R_2 = R_4 = 14k\Omega$, $R_5 = R_{12} = R_{19} = R_{24} = 100k\Omega$, $R_6 = R_7 = 10k\Omega$, $R_8 = 4k\Omega$, $R_{10} = 220k\Omega$, $R_9 = R_{11} = 22k\Omega$, $R_{13} = 20k\Omega$, $R_{14} = 10k\Omega$, $R_{15} = 110k\Omega$, $R_{16} = R_{17} = R_{18} = 10k\Omega$, $R_{20} = 60k\Omega$, $R_{21} = 16k\Omega$, $R_{22} = R_{23} = R_{25} = R_{26} = 10k\Omega$, $R_{27} = 25k\Omega$, $C_1 = C_2 = C_3 = C_4 = 1\mu F$. The parameter variable r of system (6) is changed by adjusting the resistor R_{28} , and obey the following relations:

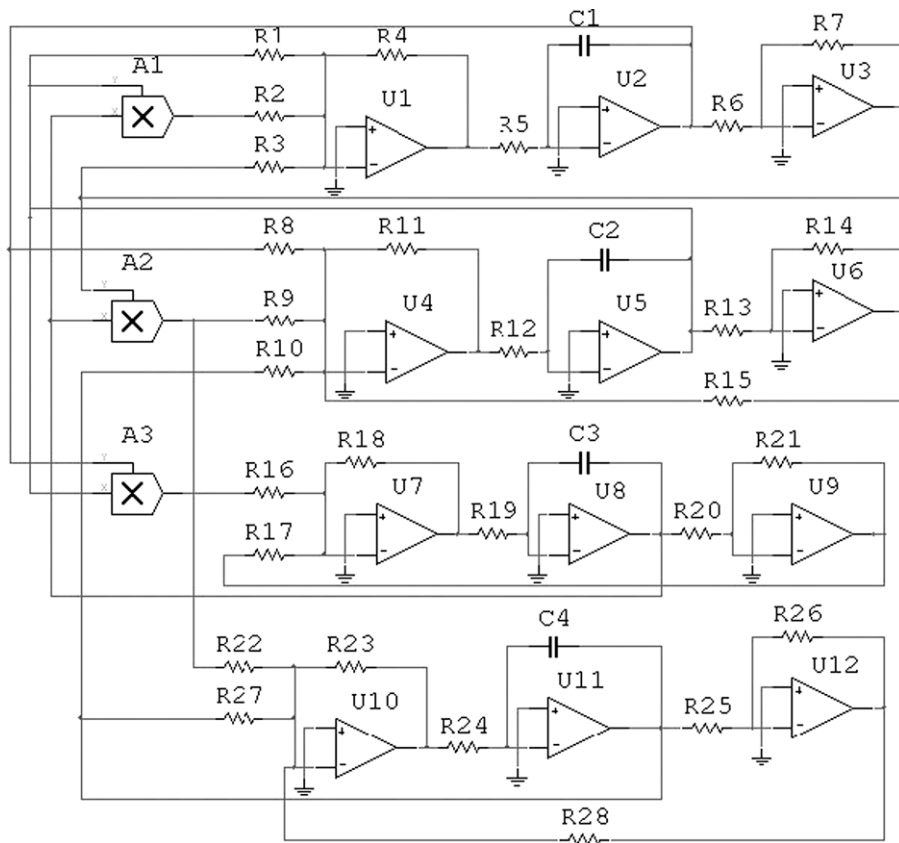


Fig. 5. Circuit diagram for system (6).

$$r = \frac{R_{23}}{R_{27}R_{24}C_4} - \frac{R_{23}R_{26}}{R_{28}R_{25}R_{24}C_4} = 4 - \frac{100}{R_{28}}, \quad (6.25k\Omega \leq R_{28} \leq 100k\Omega). \quad (8)$$

Thus, the circuit Eq. (7) is equivalent to system (6) with $a = 35, b = 8/3, c = 55, -12 \leq r \leq 3$. The supplies of all active devices are $\pm 40V$ and the initial voltages of C_1, C_2, C_3, C_4 are random. With Multisim 7.0, we obtain the experiment observations of system (6) as Figs. 6 and 7.

From Figs. 6 and 7, it can be seen that chaos and hyperchaos can be obtained by simply changing the value of R_{28} . As compared with Fig. 4(a) and (b), a good qualitative agreement between the numerical simulation and the Multisim results of the hyperchaotic circuit is confirmed.

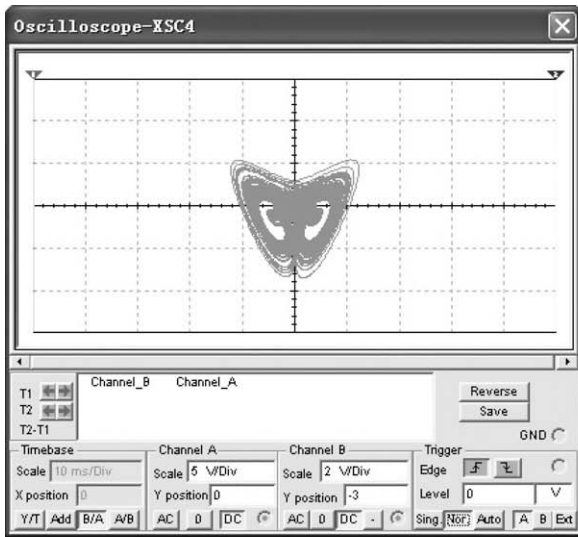
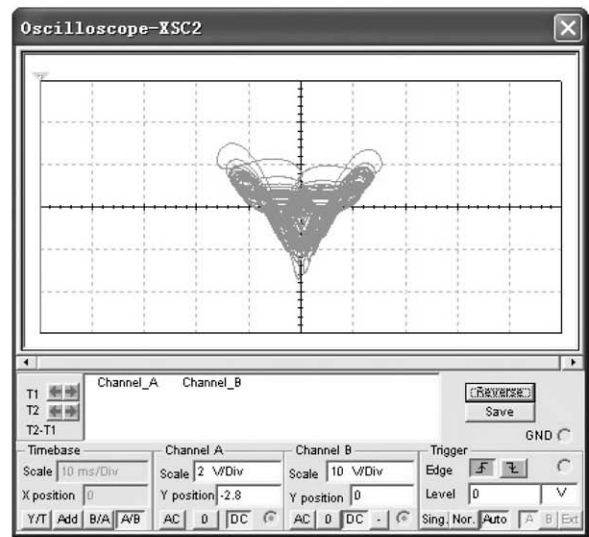
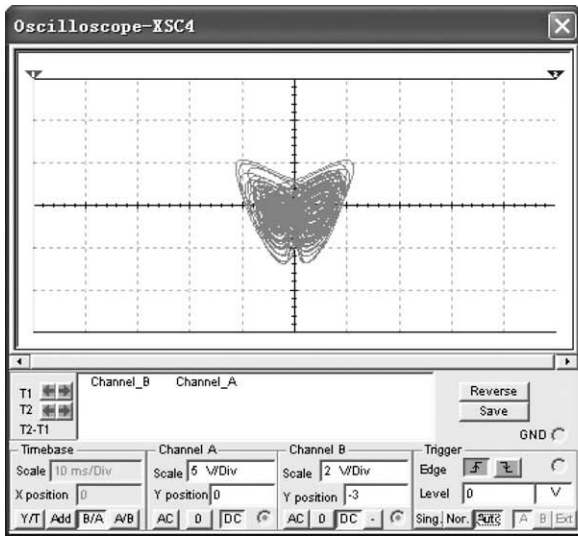
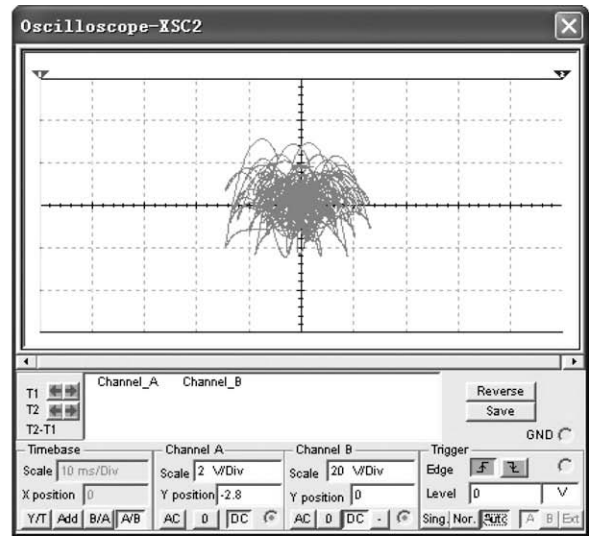
5. Discussion

It is technically challenging to construct a new hyperchaotic system based on the three-dimensional autonomous chaotic system, due to hyperchaos complicated chaotic behavior and the lack of a general fundamental theory. At present, generally, one can firstly construct a fourth-order autonomous system by introducing an additional control term, and then hyperchaotic dynamic in the fourth-dimensional autonomous system is confirmed by a combination of mathematical analysis and numerical simulations.

In this paper, based on the system (1), a new hyperchaotic system is generated by using nonlinear feedback control. In this design, for the system (1), we grope the characteristic on how to build the control law so that hyperchaos be observed. It is that hyperchaos is easily generated from the system (1), if the designed control term contains the term of the differential equation which the control term is added to. A more detailed interpretation about the hint is as below:

In the paper, for the system (1), we have done the transformation, as follows:

$$\begin{cases} \dot{x} = a(y - x) + yz \\ \dot{y} = cx - y - xz + w \\ \dot{z} = xy - bz \end{cases} \Rightarrow \begin{cases} \dot{x} = a(y - x) + yz \\ \dot{y} = cx - y - xz + w \\ \dot{z} = xy - bz \\ \dot{w} = -xz + rw \end{cases}. \quad (9)$$

(a) x - z Plane(b) w - z Plane**Fig. 6.** The experiment observations of system (6) with $R_{28} = 6.45161(\text{k}\Omega)$ ($r = -11.5$), chaotic.(a) x - z Plane(b) w - z Plane**Fig. 7.** The experiment observations of system (6) with $R_{28} = 40(\text{k}\Omega)$ ($r = 1.5$), hyperchaotic.

In the transformation (9), the control term w is added to the second differential equation and is designed as $\dot{w} = -xz + rw$. It can be seen that the control term w contains the term xz of the second differential equation.

Based on the system (1), according to the above characteristic of building the control law, we have designed two new control terms w_1 and w_2 , corresponding two new hyperchaotic systems are gained:

$$\begin{cases} \dot{x} = a(y - x) + yz \\ \dot{y} = cx - y - xz + w_1 \\ \dot{z} = xy - bz \\ \dot{w}_1 = -63x + r_1 w_1 \end{cases}, \quad (10)$$

and

$$\begin{cases} \dot{x} = a(y - x) + yz \\ \dot{y} = cx - y - xz + w_2 \\ \dot{z} = xy - bz \\ \dot{w}_2 = -x - xz + r_2 w_2 \end{cases} \quad (11)$$

From system (10) and system (11), it can be seen that the control terms w_1 and w_2 are added to the second differential equation, respectively. The control term w_1 contains the term x of the second differential equation and w_2 contains a mixture of the terms x and xz of the second differential equation.

According to the method presented by Ramasubramanian and Sriram [26], we compute the Lyapunov exponents spectrums of two new systems, respectively. We obtain that when $-0.7 < r_1 \leq 1.8$ is satisfied, $\lambda_1, \lambda_2 > 0$, $\lambda_3 < 0, \lambda_4 = 0$, the new system (10) is hyperchaotic, and the new system (11) is hyperchaotic, when $0.5 < r_2 \leq 3.4$.

On the other hand, the characteristic of building the control law is seen in literatures [13,14,27–29], too. For this characteristic, we have no corresponding theory prove. But, it may be the inspiration to the formation and perfection of the general control law. We will further study the control law on generating hyperchaos from three-dimensional autonomous chaotic system.

6. Conclusion

This paper has introduced a novel four-dimensional continuous autonomous hyperchaotic system. The complex dynamics have also been explored in detail, including various periodic, chaotic and hyperchaotic motions, by means of Lyapunov exponent spectrum and bifurcation diagrams. Moreover, it is implemented via a designed circuit and tested experimentally with Multisim. The Multisim results of the hyperchaotic circuit were well agreed with the simulation results. For the system (1) in this paper, we give some hints on how to build the control law so that hyperchaoticity be generated. It may be the inspiration to the formation and perfection of the general control law. Since the new hyperchaotic system has more complex dynamical behaviors than the normal chaotic systems, it is believed that the system will have broad applications in various chaos-based information systems.

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